

GJR-GARCH(1,1,1) Model

Portfolio VaR Engine — Model Documentation

Risk Modelling — Trade Republic

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1 Overview

The GJR-GARCH model (Glosten, Jagannathan, and Runkle, 1993) extends the standard GARCH(1,1) specification with an asymmetric term that allows negative return shocks to have a larger impact on future volatility than positive shocks of the same magnitude — the so-called *leverage effect*. This is especially relevant for equity portfolios.

Engine key: 'gjr'

Sources:

- `risk_models/garch_models.py`, lines 38–54 (`garch_gjr`)
- `risk_models/garch_models.py`, lines 58–131 (`calculate_var_garch`)
- `risk_analytics/risk_engines.py`, lines 135–152 (`engine dispatch`)

Library: `arch` (Kevin Sheppard)

2 Volatility Model

2.1 GJR-GARCH(1,1,1) Specification

Let r_t denote the portfolio return at time t . The model assumes:

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad z_t \stackrel{i.i.d.}{\sim} D(0, 1) \quad (1)$$

where $D(0, 1)$ is a standardised distribution (Normal or Student- t), and σ_t^2 follows:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbb{1}_{\{\varepsilon_{t-1} < 0\}} + \beta \sigma_{t-1}^2 \quad (2)$$

- $\omega > 0$: long-run variance intercept
- $\alpha \geq 0$: ARCH effect (symmetric impact of shocks)
- $\gamma \geq 0$: asymmetry coefficient (leverage effect; positive shocks have impact α , negative shocks have impact $\alpha + \gamma$)
- $\beta \geq 0$: GARCH persistence
- Stationarity requires: $\alpha + \frac{1}{2}\gamma + \beta < 1$

2.2 ARCH library call

The model is fitted via:

```
arch_model(rets_rescaled, p=1, q=1, o=1, vol='GARCH', dist=dist)
```

where `o=1` adds the single asymmetric (GJR) term, `p=1` is the GARCH lag order, and `q=1` is the ARCH lag order. Returns are rescaled by a factor (default 1000) before fitting for numerical stability.

2.3 EWMA Pre-weighting (optional)

Before fitting, an optional EWMA decay can be applied to down-weight older observations:

$$\tilde{r}_t = \text{EWMA}_\lambda(r_t), \quad \lambda \in (0, 1] \quad (3)$$

With $\lambda = 1$ (default for GARCH engines) no decay is applied.

3 VaR and CVaR Calculation

3.1 One-Step Forecast

After fitting, a one-step-ahead forecast of conditional mean and variance is generated using `ARCHModelResult.forecast()`:

$$\hat{\mu}_{t+1} = \hat{\mu}, \quad \hat{\sigma}_{t+1}^2 = \text{forecast variance} \quad (4)$$

The forecast uses 1000 simulations internally for the variance path (`simulations=1000`).

3.2 Parametric VaR (without FHS)

Value-at-Risk:

$$\widehat{\text{VaR}}_\alpha = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \cdot q_\alpha^D \quad (5)$$

where q_α^D is the α -quantile of the fitted distribution D .

For Normal innovations: $q_\alpha^N = \Phi^{-1}(\alpha) < 0$.

For Student- t innovations: $q_\alpha^t = t^{-1}(\alpha; \hat{\nu}) < 0$.

CVaR — Normal distribution:

$$\widehat{\text{CVaR}}_\alpha^N = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \cdot \frac{\phi(\Phi^{-1}(\alpha))}{\alpha} \quad (6)$$

CVaR — Student- t distribution:

$$\widehat{\text{CVaR}}_\alpha^t = -\hat{\mu}_{t+1} - \frac{\hat{\sigma}_{t+1}}{\alpha(1-\hat{\nu})} \left(\hat{\nu} - 2 + [t^{-1}(\alpha; \hat{\nu})]^2 \right) \cdot f_t(t^{-1}(\alpha; \hat{\nu}); \hat{\nu}) \quad (7)$$

where $\hat{\nu}$ is the MLE-estimated degrees-of-freedom, $t^{-1}(\alpha; \nu)$ is the Student- t quantile function, and $f_t(\cdot; \nu)$ is its PDF.

3.3 Filtered Historical Simulation (FHS)

When `fhs=True`, the parametric distribution is replaced by the empirical distribution of standardised residuals:

$$\hat{z}_t = \frac{r_t - \hat{\mu}}{\hat{\sigma}_t} \quad (8)$$

VaR quantiles are then taken from the empirical distribution of $\{\hat{z}_t\}$:

$$q_\alpha^{\text{FHS}} = Q_{1-\alpha}(\{\hat{z}_t\}) \quad (9)$$

(Note: the upper tail quantile $Q_{1-\alpha}$ is used to match the sign convention for losses.)

FHS retains empirical tail behaviour while using GARCH-filtered volatility, combining the strengths of both parametric and non-parametric approaches.

3.4 Holding-Period Scaling

$$\widehat{\text{VaR}}_{\alpha}^{(H)} = \widehat{\text{VaR}}_{\alpha}^{(1)} \cdot \sqrt{H} \quad (10)$$

4 Parameters

Parameter	Type	Default	Description
window	int	250	Lookback window
rescale	float	1000	Rescaling factor for returns (numerical stability)
decay	float	1	EWMA pre-weighting ($\lambda = 1$: no decay)
ftol	float	1e-6	Optimizer convergence tolerance
max_iter	int	150	Maximum optimizer iterations
fhs	bool	False	Use Filtered Historical Simulation
distr	str	'Normal'	Innovation distribution: 'Normal' or 't'
qtls	list	[0.05, 0.01, 0.001]	Tail probability levels
holding_period	int	1	Forecast horizon in days

5 Advantages and Limitations

Advantages	Limitations
Captures volatility clustering	Computationally heavier than EWMA/HS
Models leverage effect explicitly	Requires sufficient history for MLE convergence
Parametric tail estimation	Stationarity constraints may bind
FHS option for non-parametric tails	Square-root scaling is approximate

6 Usage Example

```
from risk_analytics import risk_engines as re

var_gjr = re.portfolio_var(
    rets_orig = rets,
    wgts      = weights,
    engine    = 'gjr',
    fmt_engine = {
        'window'      : 250,
        'rescale'     : 1000,
        'decay'       : 1,
        'ftol'        : 1e-6,
        'max_iter'    : 150,
        'fhs'         : False,
        'distr'       : 't',
        'qtls'        : [0.05, 0.01, 0.001],
        'holding_period' : 1,
    }
)
```

7 References

Glosten, L.R., Jagannathan, R., and Runkle, D.E. (1993). *On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks*. Journal of Finance, 48(5), 1779–1801.